Simulation Exam Project

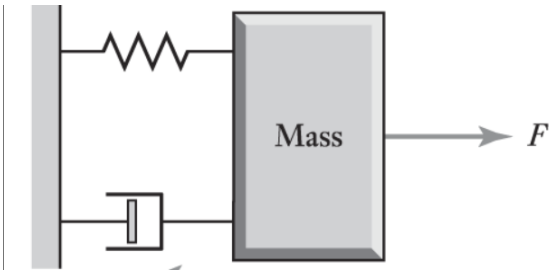
Please complete the following exam project by answering as many questions as possible. In answering all the questions, provide as many details, graphs and pictures.

Save the document as a PDF naming it starting with your name and surname. Also, include your name and surname in the document itself. Email the complete document to me at: Alfred.barnard@srh.de

The document should reach me before **23:59** hours on **Saturday the 02 April**.

Question 1

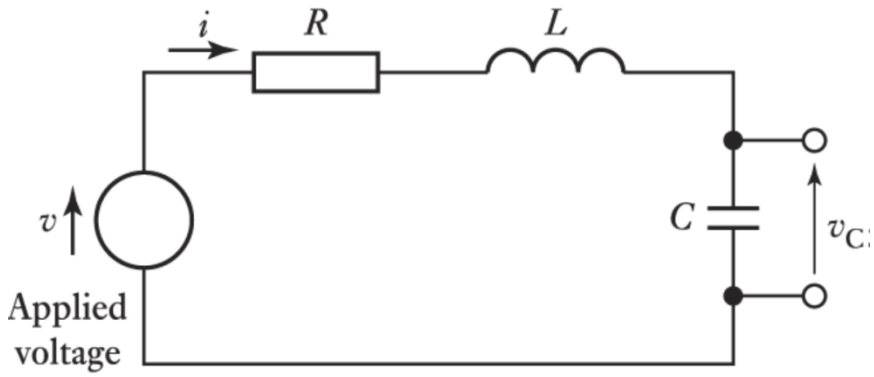
For the following system:



The spring constant is 1 N/m, the damper has a constant of 2 N.m/s and the mass weighs 1 kg.

1. Write down the differential equation.
2. Is this system underdamped, critically damped or overdamped?
3. Create a transfer function model of the system. Show me the function
4. Simulate it with a step input over a time range of 10 seconds. Show me the Simulink diagram and output figure.
5. What transfer function would you add to your model to make it unstable?

Question 2



The inductance value is 6 Henry, the resistor is 30 Ohms and the capacitor has a value of 0.01 Farad.

1. Write down the differential equation for the voltage across the capacitor.
2. Is this system underdamped, critically damped or overdamped?

3. Create a transfer function model of the system, your input is voltage and output is the voltage across the capacitor.
4. Simulate it with a step input over a time range of 10 seconds. Show me the Simulink diagram and output figure.
5. What transfer function would you add to your model to make it unstable?

Question 3

Design a low pass filter that has a cut-off frequency of 5 Hz by using only resistive and capacitive elements

1. What does a low pass filter do?
2. What are the values for the resistor and capacitor?
3. Show me the Transfer function.
4. Input a sinewave to the system that has a frequency of 100Hz, show me the Simulink model and the output scope.

Question 4

Consider the following transfer function:

$$G(s) = \frac{s^2 - 4s + 2}{s^2 + s - 6}$$

1. What are the poles and zeros for this transfer function?
2. What information does that provide about my system?
3. How can I affect the behaviour of this system?
4. Simulate the above system. Show me the Simulink diagram and output figure.

Question 5

Consider the following characteristics of a DC motor:

- (J) moment of inertia of the rotor 0.01 kg.m^2
- (b) motor viscous friction constant 0.1 N.m.s
- (Ke) electromotive force constant 0.01 V/rad/sec
- Kt) motor torque constant 0.01 N.m/Amp
- (R) electric resistance 1 Ohm
- (L) electric inductance 0.5 H

1. Create a transfer function from these characteristics. Show the starting equation and then the final transfer function. Input should be voltage and output is angular velocity.
2. Simulate the system with a step input of 10. Show the graph and Simulink diagram.
3. How fast is the motor turning during its steady-state?
4. Simulate the system, with an input alternating between 10V and -10V every 2 seconds. Show me the Simulink diagram and output figure.

Question 6

Consider the following transfer function for robot force control:

$$G(s) = \frac{F(s)}{T(s)} = \frac{K(s+2.5)}{(s^2+2s+2)(s^2+4s+5)}$$

$\tau(t)$ torque input $f(t)$ force output

1. Simulate the system with two different values for K. K = 1 and K = 0.5. Show me the Simulink diagram and output figure.
2. Is this system underdamped, critically damped or overdamped?

Question 7

Consider the following transfer function:

$$G(s) = \frac{1}{s^3 + 7.5s^2 + 11s - 7.5}$$

1. Simulate the transfer function. Show me the Simulink diagram and output figure.
2. Describe its resulting output.
3. What is causing this behaviour of the system?
4. By using simulation design a PID controller. This system should have a negative feedback loop and use a PID gain formula. The PID controller should control the system so that the input equals the steady-state output. Describe the process that you are following and show me the final Simulink model and the output.